

Setup Guide: Object Retrieval

Time Allocation: 90 Minutes

Challenge Overview

The robot must pick up a designated tennis ball using a claw attachment and transport it across the room.

Environment & Arena Setup

- **Arena Layout:** Clear a flat, open floor space measuring approximately 10x10 feet (or adjust based on classroom size).
- **Start and Target Zones:** Mark a designated **Start Zone** (e.g., 2x2 feet square using colored tape) in one corner of the room. Mark a corresponding **Target Drop-off Zone** diagonally across the room.
- **Object Placement:** Place a single standard **tennis ball** precisely in the center of the arena or at a predefined waypoint between the start and target zones.
- **Obstacles (Optional):** For advanced groups, add minor physical barriers (like low cardboard blocks) along the path to encourage obstacle avoidance or path-planning algorithms.
- **Safety Boundary:** Use cones or tape to create a perimeter boundary to keep the robots contained within the designated challenge area and protect surrounding classroom furniture.